BICYCLE COMPANY SALES

Data Analytics Project — Comprehensive Documentation

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Tool: Microsoft Power BI | Period: 2024–2026

Report Generated: February 2026

$3.51M	$45.81K	7,000	1.31%
Gross Revenue	Net Profit	Units Sold	Profit Margin

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1. Project Overview

This document provides a comprehensive analytical breakdown of the Bicycle Company Sales Data Analytics Portfolio Project. The project was built using Microsoft Power BI and covers end-to-end data analysis across five relational tables, seven derived analytical measures, and four interactive dashboards. The purpose is to demonstrate applied data analytics competencies, including data modelling, measure derivation using DAX (Data Analysis Expressions), dashboard design, and business intelligence storytelling.

This project is published as part of a professional LinkedIn Data Analytics Portfolio and represents a comprehensive analytics workflow, spanning from raw transactional data to actionable executive insights.

1.1 Project Objectives

- Transform raw bicycle company transactional data into a structured, interactive Power BI data model.
- Derive key business performance measures using DAX formulas.
- Build multi-layered dashboards supporting Country, Category, Product, and Salesperson analysis.
- Generate data-driven business insights and strategic recommendations for executive decision-making.
- Produce a portfolio-quality deliverable suitable for LinkedIn and professional data analytics presentations.

1.2 Tools & Technologies

- Microsoft Power BI Desktop — Primary analytics and dashboard tool
- DAX (Data Analysis Expressions) — Measure derivation and calculations
- Power Query — Data transformation and loading
- Microsoft PowerPoint — Analysis slide deck
- Microsoft Word — Project documentation report

2. Data Model & Table Documentation

The data model consists of five interrelated tables forming a Star Schema structure, a widely adopted design pattern in business intelligence and data warehousing. This structure optimises query performance and supports efficient filtering and aggregation across dimensions.

2.1 Salesperson Table

The Salesperson table is a dimension table containing identity and role information for all sales representatives. It has 18 rows, each uniquely identifying a salesperson.

Row Count: 18 rows | Purpose: Dimension Table — Sales Rep Identity

Column Name	Data Type	Description	Example Value	Notes
EmployeeKey	Integer	Primary key – unique identifier for each employee	1, 2, 3…	Joins to Sales table
EmployeeID	String	HR system employee reference ID	E001, E002…	Internal HR reference
Salesperson	String	Full name of the sales representative	Tsvi Reiter	Used in visuals
Title	String	Job title or role designation	Sales Rep	Descriptive only
UPN (Email)	String	User Principal Name / email address	tsvi@company.com	System login reference

2.2 Date Table

The Date table is a time intelligence dimension table providing date hierarchy support. It enables time-based filtering, year-over-year analysis, and monthly trend visualisation.

Row Count: 1,461 rows (4 years of daily data) **| Purpose:** Dimension Table — Calendar / Time Intelligence

Column Name	Data Type	Description	Example Value	Notes
Date	Date	Full calendar date — primary key	01/01/2022	Marked as date table in Power BI
Year	Integer	Calendar year extracted from Date	2022, 2023	Used for YoY comparisons
Month	String	Month name (short or full form)	Jan, February	Used in charts/slicers
MonthNum	Integer	Numeric month for correct sorting	1 through 12	Critical for sort order

2.3 Products Table

The Products table is a dimension table describing all bicycle company products. With 397 rows, it spans all product variants across four categories.

Row Count: 397 rows | Purpose: Dimension Table — Product Master

Column Name	Data Type	Description	Example Value	Notes
ProductKey	Integer	Primary key – unique product ID	1, 2, 3…	Joins to Sales table
Product	String	Product name or SKU description	Mountain Bike Pro	Displayed in visuals
Standard Cost	Currency	Manufacturing / acquisition cost per unit	$420.50	Used in cost calculations
Colour	String	Product colour variant	Red, Blue, Silver	Product attribute
Subcategory	String	Sub-level product grouping	Mountain Bikes	Hierarchical grouping
Category	String	Top-level product category	Bikes, Accessories	4 categories total
Background Colour Format	String	Hex colour code for conditional formatting	#0D1B2A	Dashboard formatting
Font Colour Format	String	Hex colour code for font formatting	#FFFFFF	Dashboard formatting
Category ID	Integer	Numeric ID for category	1, 2, 3, 4	Lookup reference
Subcategory ID	Integer	Numeric ID for subcategory	1 through 38	Lookup reference

2.4 Region Table

The Region table is a geographic dimension table mapping sales territories to countries and regional groups. It supports country-level and regional filtering in the Country Performance Dashboard.

Row Count: 10 rows | Purpose: Dimension Table — Geography / Territory

Column Name	Data Type	Description	Example Value	Notes
SalesTerritoryKey	Integer	Primary key – territory identifier	1, 2, 3…	Joins to Sales table
Region	String	Sales region or territory name	Northwest, Central	Sub-country grouping
Country	String	Country name where sales occurred	United States, Canada	6 countries total
Group	String	Macro geographic grouping	North America, Europe	Continental grouping

2.5 Sales Table (Fact Table)

The Sales table is the central fact table of the data model. It records every individual sales transaction and stores the quantitative measures (foreign keys and numeric values) that drive all KPI calculations. With 3,616 rows, it represents the company's complete transactional record.

Row Count: 3,616 rows | Purpose: Fact Table — Transactional Sales Records

Column Name	Data Type	Description	Example Value	Notes
SalesOrderNumber	String	Unique sales order reference	SO-00001	Transaction identifier
OrderDate	Date	Date of sale transaction	15/03/2022	Links to Date table
ProductKey	Integer	Foreign key to Products table	42	Links product to sale
ResellerKey	Integer	Reseller/distributor identifier	501, 502…	Channel reference
EmployeeKey	Integer	Foreign key to Salesperson table	3	Links rep to sale
SalesTerritoryKey	Integer	Foreign key to Region table	2	Links region to sale
Quantity	Integer	Number of units sold in transaction	2, 5, 10	Volume metric
Unit Price	Currency	Selling price per unit at transaction	$534.35	Revenue per unit
Cost	Currency	Cost price per unit at transaction	$527.37	Cost per unit
Total Sales	Currency	Quantity × Unit Price (transaction total)	$1,068.70	Key revenue column

3. Derived Analytical Measures (DAX)

Seven key performance indicators (KPIs) were derived from the raw table data using DAX (Data Analysis Expressions) formulas in Power BI. These measures form the backbone of all dashboard cards, charts, and analytical visualisations. Below is a detailed explanation of each measure, its formula logic, and its business significance.

Measure Name	DAX Formula Logic	Dashboard Label	Business Significance
Total Sales	SUM(Sales[Total Sales])	Gross Revenue	Total monetary value of all sales transactions. Primary top-line revenue metric.
Total Cost	SUMX(Sales, Sales[Quantity] × Sales[Cost])	Cost of Goods Sold (COGS)	Total direct cost of all units sold. Key input for margin and profitability analysis.
Total Profit	[Total Sales] – [Total Cost]	Net Profit	Absolute profit generated after deducting cost of goods. Shows bottom-line performance.
Profit Margin	DIVIDE([Total Profit], [Total Sales])	Profit Margin %	Percentage of revenue retained as profit. Critical for pricing strategy evaluation.
Total Quantity	SUM(Sales[Quantity])	Units Sold	Total number of product units sold across all transactions. Volume metric.
Average Selling Price	AVERAGE(Sales[Unit Price])	Avg Sell Price	Mean selling price per unit. Indicates product pricing tier and market positioning.
Avg Cost Per Unit	AVERAGE(Sales[Cost])	Avg Cost/Unit	Mean cost per unit sold. Paired with ASP to evaluate per-unit profitability.

3.1 KPI Naming Recommendations for Dashboard

Professional analytics dashboards use standardised financial terminology. Below are the recommended professional titles for the three primary KPI cards:

- **Total Sales → Gross Revenue:** Industry-standard financial term for total top-line sales. Communicates financial literacy to stakeholders.
- **Total Profit → Net Profit / Net Income:** The globally recognised term for bottom-line profit performance. 'Operating Profit' is an alternative if taxes are excluded.
- **Total Cost → Cost of Goods Sold (COGS):** Standard P&L terminology for direct production/acquisition costs. Positions the dashboard within a formal financial reporting framework.

4. Dashboard Analysis — Slide-by-Slide Breakdown

4.1 Country Performance Dashboard

Overview

The Country Performance Dashboard provides a geographic breakdown of business performance across six countries: United States, Canada, France, United Kingdom, Australia, and Germany. Interactive filter buttons allow stakeholders to isolate individual country performance or view aggregated totals across all markets.

Key Metrics (All Countries)

KPI	Value	Professional Title
Total Sales	$3.51M	Gross Revenue
Total Profit	$45.81K	Net Profit
Total Quantity	7,000 units	Units Sold
Total Cost	$3.46M	Cost of Goods Sold (COGS)
Average Selling Price	$534.35	Average Selling Price (ASP)
Average Cost Per Unit	$527.37	Average Unit Cost
Profit Margin	1.31%	Net Profit Margin

Dashboard Analysis

The United States is the clear market leader, contributing 65.07% of total profit and generating over $2 million in gross revenue — more than double all other markets combined. This dominant position reflects the maturity of the US bicycle market and potentially a well-established distribution network.

Canada ranks as the second most important market at 27.3% of total profit, making it a strategic priority for growth investment. The combined US-Canada performance accounts for over 92% of total company profit, revealing a high geographic concentration risk.

France, United Kingdom, Australia, and Germany collectively represent just 7.5% of total profit. While these numbers are modest, they signal significant untapped market potential, particularly in European markets where cycling adoption has grown substantially.

The 1.31% profit margin is critically low. With only a $6.98 spread between the average selling price ($534.35) and average cost ($527.37), the company operates with minimal pricing buffer. Any cost increase — logistics, materials, labour — could push the business into losses.

Strategic Business Decisions

- Retain & Grow the US Market: Invest in customer retention programs, premium product lines, and digital marketing to protect the dominant US market position.
- Accelerate Canadian Growth: Allocate incremental marketing budget to Canada. A targeted 10% sales growth would add over $350K in revenue.
- Develop a European Expansion Strategy: Conduct market feasibility studies for France and UK. Appoint regional sales managers or channel partners.
- Margin Improvement Initiative: Implement cost reduction measures — supplier renegotiations, logistics optimisation, and a pricing power review to achieve 3-5% net margin within 12 months.

4.2 Product Category Performance Dashboard

Overview

The Product Category Performance Dashboard segments business performance across four product categories: Bikes, Components, Clothing, and Accessories. The dashboard features a funnel chart showing sales volume by category and a pie/donut chart showing profit contribution by category.

Category Performance Breakdown

Category	Total Sales	Profit Contribution	Assessment
Bikes	$2.91M	83.28%	Core Revenue Driver
Components	$0.56M	12.10%	Strong Support Category
Clothing	$0.03M	4.62%	Growth Opportunity
Accessories	$0.01M	~0.02%	Severely Underdeveloped

Funnel Chart Explanation

The funnel chart visually represents the steep sales cascade across categories. The widest bar at the top represents Bikes ($2.91M), progressively narrowing through Components ($0.56M), Clothing ($0.03M), and Accessories ($0.01M). This chart is particularly powerful for communicating category concentration and the sharp drop-off in revenue contribution below the Bikes category.

Dashboard Analysis

Bikes represent 83.28% of all profit — positioning the company as essentially a bicycle business with marginal product diversification. This extreme concentration in a single category presents both a strength (market expertise) and a vulnerability (single-product dependency risk).

Components at 12.1% represent a meaningful secondary revenue stream, likely driven by replacement parts and upgrade demand from existing bike owners. The relationship between Bikes and Components presents a cross-sell opportunity.

Clothing at 4.62% and Accessories under 1% are severely underperforming for a bicycle company. Branded cycling apparel and accessories typically command high margins and strong brand affinity — their minimal contribution suggests inadequate product range, pricing, or marketing investment.

Strategic Business Decisions

- Expand the Premium Bike Range: Invest in e-bikes, performance road bikes, and urban commuter bikes. Forecast an additional 15% revenue uplift from premium segment expansion.
- Component Bundling Strategy: Create pre-configured bike + components packages to increase average basket size and lift component revenue above 20%.
- Clothing & Accessories Revitalisation: Commission a branded cycling apparel line. Set a combined Clothing + Accessories revenue target of 10% within 18 months.
- Category Concentration Risk Policy: Establish a board-level KPI that no single category should exceed 70% of total revenue within 3 years.

4.3 Individual Product Performance Dashboard

Overview

The Individual Product Performance Dashboard provides the most granular view, enabling drill-down analysis at the individual product level across all 397 SKUs. The advanced product slicer allows users to select individual products or groups of products for targeted comparison.

Dashboard Analysis

Across 397 products, the Bikes category contributes $2,908,497.81 (83% of $3,505,340.43 total revenue). Components generate $558,467.71 (16%), Clothing $31,308.59 (0.9%), and Accessories just $7,066.32 (0.2%).

The product slicer functionality is critical for identifying top-performing and underperforming SKUs. Products visible in the initial view include All-Purpose Bike Stand, AWC Logo Cap, Bike Wash Dissolver, Cable Lock, Chain, Classic Vest variants, and Fender Set, largely accessories/clothing items, which aligns with their low aggregate revenue contribution.

Strategic Business Decisions

- SKU Rationalisation: Use the product slicer to identify the bottom 20% of products by revenue. Discontinue or reposition underperforming SKUs to reduce inventory carrying costs.
- Accessories Upsell Programme: Launch a 'Complete Your Ride' point-of-sale accessories bundle. Target $50,000 in accessories revenue within 6 months.
- Top Bike Protection: Identify the top 10 best-selling bike models and establish safety stock levels and supplier Service Level Agreements (SLAs) to prevent stockouts.
- Product Performance Dashboard: Share this dashboard with the procurement team for data-driven purchasing decisions based on sales velocity.

4.4 Sales Performance by Salesperson Dashboard

Overview

The Sales Performance by Salesperson Dashboard ranks all 18 sales representatives by total sales performance. Interactive salesperson selection enables individual and grouped performance analysis. A pie chart displays each salesperson's sales mix by product category.

Salesperson Rankings

Rank	Salesperson	Estimated Sales	Performance Band
1	Tsvi Reiter	~$475K	Elite
2	Michael Blythe	~$440K	Elite
3	Jillian Carson	~$410K	Elite
4	Linda Mitchell	~$360K	High Performer
5	José Saraiva	~$300K	High Performer
6	Jae Pak	~$265K	High Performer
7	Shu Ito	~$245K	Standard Performer
8	Garrett Vargas	~$215K	Standard Performer
9	David Campbell	~$210K	Standard Performer

10-14	Pamela Ansman-Wolfe et al.	$100K-$195K	Developing
15-18	Amy Alberts, Syed Abbas et al.	<$100K	Underperforming

Dashboard Analysis

The sales team exhibits a classic Pareto distribution (80/20 rule). The top 3 performers — Tsvi Reiter, Michael Blythe, and Jillian Carson — collectively drive approximately 38-40% of total company revenue. This pattern is common in B2B and B2C sales environments but signals a talent concentration risk.

The category mix (Bikes 82.97%, Components 15.93%, Clothing 0.89%, Accessories <0.2%) mirrors the overall company portfolio, suggesting that no individual salesperson has meaningfully differentiated their selling mix toward higher-margin or underserved categories.

The bottom 4 performers (Amy Alberts, Stephen Jiang, Rachel Valdez, Syed Abbas) each recorded less than $100K in sales. This could reflect territory imbalances, product knowledge gaps, or performance management issues requiring investigation.

Strategic Business Decisions

- **Elite Performer Retention:** Create a tiered commission structure and annual 'Sales Champion' award to retain the top 3-5 performers who drive 40% of revenue.

- **Mentorship Programme:** Pair the top 3 performers with the bottom 5 as mentors. Set structured 90-day improvement plans with measurable sales targets.

- **Territory Rebalancing:** Audit territory assignments for the bottom performers. Reassign territories based on market opportunity data rather than historical allocation.

- **Category Diversification Incentive:** Introduce bonus incentives for selling Clothing and Accessories to drive portfolio-wide sales growth beyond the Bikes-only focus.

- **Real-Time Salesperson Dashboard:** Provide each rep access to their personalised Power BI performance view for self-directed accountability and goal tracking.

5. Executive Summary & Overarching Business Insights

5.1 Business Performance Summary

The Bicycle Company achieved $3.51M in Gross Revenue across the analysis period, generating a Net Profit of $45.81K at a 1.31% Net Profit Margin. While the revenue figure is substantial, the razor-thin margin is the single most critical issue requiring immediate attention. The $6.98 gap between average selling price ($534.35) and average cost ($527.37) leaves the business extremely vulnerable to any cost increase or pricing pressure.

5.2 Key Strengths

- Strong US market dominance (65% of profits) with established customer base.
- Consistent performance of the top salesperson tier (Tsvi Reiter, Michael Blythe, Jillian Carson).
- Bikes category strength with $2.91M in sales demonstrating clear product-market fit.
- Robust Power BI data model enabling multi-dimensional, real-time performance analysis.

5.3 Critical Risks

- 1.31% profit margin offers no buffer against cost increases, supply chain disruptions, or economic downturns.
- 83% category concentration in Bikes creates dangerous single-product dependency.
- 65% geographic concentration in the US represents a market risk if US demand softens.
- Revenue heavily dependent on 3 of 18 salespeople (approximately 40% of total sales).
- Accessories ($7K) and Clothing ($31K) are severely underperforming, representing lost revenue opportunity.

5.4 Priority Strategic Recommendations

1. **Urgent — Margin Improvement:** Conduct a full pricing and cost review to raise net profit margin to a minimum 3% within 12 months.
2. **High Priority — Product Diversification:** Develop and market Clothing and Accessories lines with a target of 10% combined revenue contribution within 18 months.
3. **High Priority — Sales Team Optimisation:** Implement tiered incentive programme, mentorship pairings, and territory rebalancing.
4. **Strategic — Geographic Diversification:** Develop a Europe and Pacific market entry strategy targeting France, UK, Germany, and Australia.
5. **Ongoing — Data-Driven Culture:** Expand Power BI usage to all department heads with personalised dashboards for procurement, marketing, and regional management.

6. Project Notes & Portfolio Context

6.1 Data Analytics Skills Demonstrated

- **Data Modelling:** Star schema design with 1 fact table and 4-dimension tables.
- **DAX Measures:** 7 custom KPI measures covering revenue, cost, profit, margin, volume, and pricing.
- **Visualisation Design:** 4 interactive dashboards with slicers, bar charts, pie/donut charts, funnel charts, and KPI cards.
- **Business Intelligence Storytelling:** Converting raw data into executive-ready insights and strategic recommendations.
- **Dashboard Interactivity:** Button slicers for Country, Product Category, Individual Product, and Salesperson filtering.
- **Professional Documentation:** Comprehensive project documentation with table schema, measure logic, and business decision frameworks.

6.2 How to Use This Report

This document is intended to accompany the Power BI Dashboard and the PowerPoint Analysis Slides as part of a complete analytics portfolio submission. Readers, whether recruiters, clients, or collaborators, can use this document to understand the full analytical process, data structure, and business intelligence capability demonstrated in this project.

6.3 Document Metadata

Field	Details
Project Title	Bicycle Company Sales — Data Analytics Portfolio Project
Tool	Microsoft Power BI Desktop
Report Date	February 2026
Data Rows Analysed	3,616 sales transactions
Products Covered	397 product SKUs
Countries Analysed	6 (US, Canada, France, UK, Australia, Germany)
Salespeople Profiled	18 representatives
DAX Measures Created	7 custom KPI measures
Dashboards Built	4 interactive Power BI dashboards